Clinton Enwerem

Santa Clara, CA, United States
enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

Summary

Robotics Research Engineer with 5+ years of research experience and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration pipelines, evaluate imitation/RL policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical robots.

Skills

Robotics: cuRobo, ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, OpenVLA-7B, QLoRA, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, Gazebo, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **GPU Computing:** CUDA, Warp, MuJoCo Warp, JAX; **Perception:** FoundationPose, BundleSDF, SAM 2, SAM 3D, PCL, OpenCV; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Institute for Systems Research (ISR), University of Maryland Aug. 2021 – Present
Graduate Research Assistant *College Park, MD*
• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Trained PPO with GDF rewards in **1,024 Isaac Lab environments**: **64/64** pregrasp arrivals versus **0/64** with palm-distance rewards.
• Developed xArm7-RealHand-L6 VR teleoperation with wrist-to-Cartesian velocity mapping, relative joint-position control, grasp primitives, thumb abduction, and hand reset; integrated MoveIt Servo with Dimensional OS (DimOS) for collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
• Fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing; each of three training seeds achieved 136/200 task completions versus 129/200 without supervision, recovering seven failures without disrupting nominal successes.
• Developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. Evaluated task execution in MuJoCo.

MATRIX Lab, USM at Southern Maryland Jun. – Aug. 2022
Research Intern *California, MD*
• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Kognitive Robotics Mar. 2021 – Aug. 2021
Robotics Engineer *Lagos, Nigeria*
• Designed power and sensor–actuator interfaces in Altium; assembled a mobile robot and developed and tested ROS software for the platform.

Robotics & Artificial Intelligence Nigeria (RAIN) Jan. 2020 – Feb. 2021
Robotics Trainee *Ibadan, Nigeria*
• Developed a ROS-based quadrotor autonomy stack on a Raspberry Pi 3, integrating ORB-SLAM3 visual SLAM, an EKF fusing SLAM poses with IMU data, and a PID flight controller. The onboard controller maintained stable hover during ground-station packet loss.

Projects

ParcelStow — Execution-Timing Robustness in Imitation Learning (Paper, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Built ParcelStow, an Isaac Lab benchmark for assessing imitation-policy robustness to execution-timing variation; generated **970K** expert states (**297** demonstrations) and trained ACT, Diffusion Policy, and DAgger. Controlled timing stress tests exposed precision-insertion failures: at 2× execution speed, ACT achieved **53%** success versus **84%** for the expert.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

dexreach — GPU-Accelerated Dexterous Grasp and Motion Planning (Project) | *cuRobo, PyTorch, Isaac Lab* 2026
• Built GPU-accelerated grasp synthesis and cuRobo motion planning with friction/actuator checks and Isaac Lab validation: dual Franka-Ability completed both objects' sequences in **6/6** tested plans, with **4–19 mm** final position errors.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression." *CDC 2025*. arXiv:2506.06954.
• Additional first-author papers in control, safe learning, and robotic manipulation: IROS 2026; CDC 2026/2025/2024. Full list: Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018

Clinton Enwerem

Santa Clara, CA, United States

enwerem@umd.edu | 301-416-9430 (Mobile)

Dear Technical Recruiter,

Application: Research Scientist, Robotics

I work on dexterous manipulation, robot learning, motion planning, state estimation, and feedback control. I am an Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, advised by Dr. John Baras and Dr. Calin Belta and graduating in Fall 2026. I formulate technical questions, implement the resulting methods in Python and C++, and evaluate them in simulation and on physical arms with dexterous hands.

In ParcelStow, I built an Isaac Lab Unitree G1 and RealHand L6 grasp–reorient–insert task, collected 297 successful expert demonstrations, and trained DAgger, Diffusion Policy, and ACT. The expert and one ACT policy both reached 100% nominal success. At twice nominal speed, expert success was 84% and ACT success was 53%, a 31-percentage-point difference within the demonstrated speed range. Stage outcomes placed 35 of ACT's 47 failures during insertion.

My manipulation research also addresses uncertain contact and feedback execution. FIRMGrasp uses a CVaR wrench-space margin to rank grasps under friction uncertainty. Grasp Distance Fields use a CBF-CLF quadratic program during execution; simulated experiments lifted 46 of 50 objects, retained a median 94% of the certified grasp margin, and solved the program in 0.09 ms. In EquiDexFlow, I developed a flow-matching grasp generator and retargeted its predicted contacts to a physical LEAP Hand. I fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing. Each of three training seeds completed 136/200 tasks versus 129/200 without supervision, recovering seven failed executions without disrupting nominal successes. I developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. I evaluated task execution in MuJoCo.

I developed dexreach to connect GPU-accelerated dexterous grasp synthesis and cuRobo motion planning with Isaac Lab sequence validation, evaluating object motion, task-stage completion, and contact-loss stops.

My resume, research statement, and curriculum vitae are attached. ParcelStow code and data and models are available online. You can reach me at **enwerem@umd.edu** or **301-416-9430**.

Thank you for considering my application. I am available to start in December 2026. I look forward to discussing how my experience in robot learning, simulation, and feedback control could support your team's work.

Sincerely,

Clinton Enwerem
October 27, 2026

Clinton Enwerem

Research Statement

Santa Clara, CA, United States

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

Learning and Feedback Control for Dexterous Manipulation

My research concerns the selection and execution of manipulation actions when contact properties are uncertain and learned behavior changes with execution conditions. I develop grasp representations, state estimators, feedback controllers, and policy evaluations, and implement the software needed to study them in simulation and on physical arms and dexterous hands. My objective is to develop manipulation systems that use perception and contact feedback to decide when to continue an action, revise a grasp, or seek additional information.

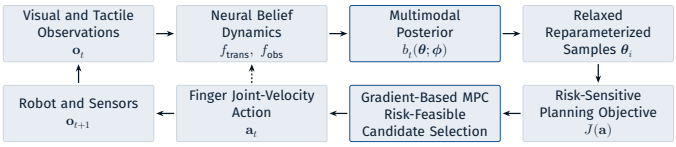


Figure 1. VNB-Grasp: Belief and Feedback Control
Visual/tactile observations update a multimodal belief for risk-sensitive action selection.

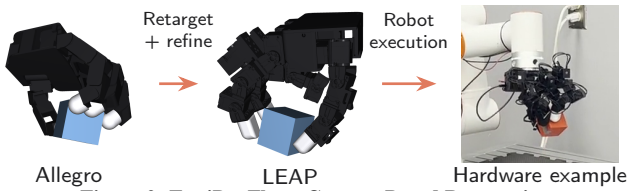


Figure 2. EquiDexFlow: Contact-Based Retargeting
Predicted contacts support retargeting from Allegro to LEAP and hardware execution.

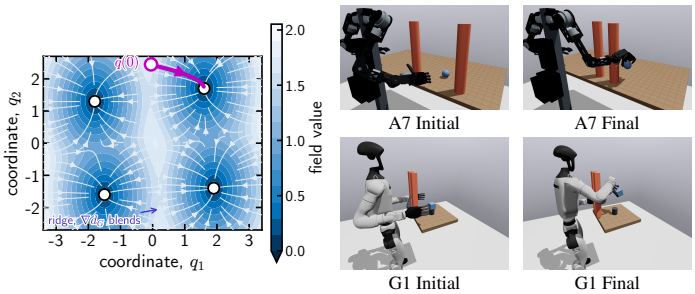


Figure 3. GDF: Field-Guided Grasp Execution
Smooth configuration-space field with initial/final A7 and G1 grasp-execution states in simulation.

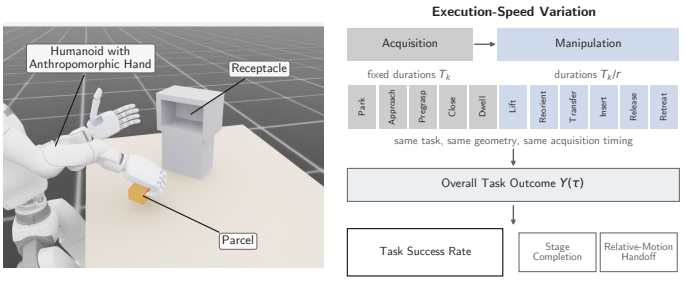


Figure 4. ParcelStow: Controlled Timing Evaluation
Acquisition timing stays fixed while manipulation durations vary; outcomes and task stages are evaluated separately.

Contact Uncertainty and Feedback Execution

Object pose, friction, and contact behavior cannot generally be determined from a single observation. My IROS 2026 work, VNB-Grasp [1] (Fig. 1), represents pose and latent contact parameters with an eight-component Gaussian mixture. Relaxed sampling supports pathwise gradients of a smooth conditional value-at-risk (CVaR) objective for action optimization. The representation retains separated hypotheses instead of collapsing uncertainty into a single Gaussian. I evaluated the method across 72 MuJoCo conditions per method, varying friction regime, risk level, randomization, and object. In physical experiments across three objects and four planar pose offsets, VNB-MPC and Gaussian MPC each succeeded in 12 of 12 trials; median episode time was 11.5 s for VNB-MPC and 14.2 s for Gaussian MPC. The experiments distinguish the effects of the belief and controller from the hardware task's success rate.

My complementary work on FIRMGrasp [2] studies grasp selection under friction uncertainty through a CVaR wrench-space margin and a probabilistic force-closure certificate. This formulation makes friction sensitivity part of grasp evaluation. FIRMGrasp and VNB-Grasp examine uncertainty before execution, when selecting a grasp, and during execution, when observations can change the estimated contact conditions.

An action also changes the information available for subsequent decisions. In Belief-Informed Hybrid Control [6], I developed a dual-control algorithm whose two-step lookahead updates the parameter belief from predicted observations before evaluating the next action. An expected-decrease constraint and a bound on cumulative slack permit informative deviations from the target set. Under the stated prediction and feasibility assumptions, the method guarantees almost-sure target-set convergence at decision times. Simulated bimanual assembly experiments examine how contact feedback changes action admissibility under friction uncertainty.

In Grasp Distance Fields [3] (Fig. 3), I use a configuration-space distance field over feasible pregrasp configurations to guide feedback grasp execution. A control-barrier-function/control-Lyapunov-function quadratic program combines progress toward the grasp with collision and workspace constraints. In simulated grasp-and-lift experiments, the controller lifted 46 of 50 objects, correctly rejecting the remaining four grasps as insufficiently stable. The median QP solve time was 0.09 ms. Drake simulation experiments additionally examine reactive execution with changing obstacles across arm–hand and humanoid models.

Learning and Evaluating Executable Skills

In EquiDexFlow [5] (Fig. 2), I developed a flow-matching grasp model predicting wrist pose, joint configuration, fingertip contacts, and forces, trained on 8,100 grasps across 81 objects. I retargeted predicted contacts to a physical LEAP Hand and demonstrated ten successful pick-and-hold executions across six objects. I also implement ROS 2 hand interfaces, tactile acquisition, VR teleoperation, and motion-planning software. My RealHand L6 software includes a C++ SocketCAN driver, joint and tactile interfaces, contact-gated control, and MoveIt 2 configuration. I verified communication, feedback, and trajectory execution on the physical hand.

A configuration-space grasp distance field can also define a learning objective. I trained PPO in 1,024 parallel Isaac Lab environments using grasp-distance-field rewards for a Unitree G1 with a dexterous hand. The policy reached the pregrasp pose in 64/64 evaluation episodes versus 0/64 with palm-distance rewards. The evaluation measures pregrasp reaching.

I built ParcelStow [4] (Fig. 4) to investigate whether nominal imitation performance persists when task execution timing changes. In Isaac Sim and Isaac Lab, I authored manipulation environments and generated 970K expert states from 297 demonstrations, then trained ACT, Diffusion Policy, and DAgger. The insertion benchmark compares policies with a scripted expert under matched execution speeds and initial conditions. At twice nominal speed, ACT succeeded in 53% of trials compared with 84% for the expert. Most ACT failures occurred during insertion. Stage-level evaluation therefore identifies a precision-manipulation limitation that aggregate nominal success does not reveal. I will test whether additional demonstrations, different policy inputs, or feedback corrections reduce the observed insertion failures.

I fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing. Each of three training seeds completed 136/200 tasks versus 129/200 without supervision, recovering seven failed executions without disrupting nominal successes. I developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. I evaluated task execution in MuJoCo.

In dexreach, I extend cuRobo with GPU-accelerated dexterous grasp synthesis, friction and actuator feasibility checks, and whole-body motion planning. An Isaac Lab validator records object motion, task-stage completion, and contact-loss stops, supporting repeatable studies of how grasp selection and motion execution affect manipulation outcomes.

Future Research

I will study how evolving pose and contact estimates affect manipulation actions, when failure analysis identifies a need for additional demonstrations, and how learned grasp proposals can be combined with constrained feedback execution. Controlled changes in sensing, contact conditions, and execution timing will separate errors in perception, action selection, and control. Repeatable comparisons with unchanged baselines will test whether each intervention improves task completion while preserving the applicable safety constraints.

Selected References

[1] C. Enwerem et al. *Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty*. IROS 2026, pp. 3117–3124. arXiv:2604.25897.
[2] C. Enwerem et al. *FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis*. Preprint, 2026. arXiv:2607.25049.
[3] C. Enwerem et al. *Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality*. Preprint, 2026. arXiv:2608.00600.
[4] C. Enwerem et al. *Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds*. Preprint, 2026. arXiv:2609.01453.
[5] C. Enwerem et al. *EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows*. Preprint, 2026. arXiv:2606.12728.
[6] C. Enwerem, S. Kemal, J. S. Baras, and C. Belta. *Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence*. Preprint, 2026. arXiv:2610.07674.

Clinton Enwerem

Academic CV

Santa Clara, CA, United States

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Research Interests

Dexterous Manipulation, Robot Learning, Motion Planning, Risk-Sensitive Control, Safety-Critical Control, and State Estimation under Multimodal Uncertainty.

Education

Ph.D., Electrical & Computer Engineering 2021–Fall 2026
University of Maryland, College Park | MD, USA
Advisors: Professor John S. Baras and Professor Calin Belta. **Admitted to Candidacy**: June 2025.
Relevant Coursework: Decision-Making for Robotics, Network Control Systems, Decision Making Under Uncertainty, Random Processes, Advanced Digital Signal Processing, Nonlinear Control Systems, Optimal Control, System Theory, Convex Optimization, Formal Methods for Control & Dynamical Systems, Multimodal Foundational Models.

B.Eng., Electrical Engineering 2014–2018
University of Nigeria, Nsukka | Enugu, Nigeria
Graduated with Highest Honors. *Emphasis*: Control Theory. *Completed coursework in engineering numerical analysis, control systems, electrical machines and drives, computer architecture, electronics, and communication systems*.

Publications

Preprints / Articles In Review

[P5] **Clinton Enwerem**, Saleh Kemal, John S. Baras, and Calin Belta. Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence, October 2026. Preprint. *2026*

[P4] **Clinton Enwerem**, John S. Baras, and Calin Belta. Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds. arXiv preprint, September 2026. arXiv:2609.01453 Code Data and Models. *2026*

[P3] **Clinton Enwerem**, John S. Baras, and Calin Belta. Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality, August 2026. arXiV Link Project Page. *2026*

[P2] **Clinton Enwerem**, John S. Baras, and Calin Belta. FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis, July 2026. arXiV Link. *2026*

[P1] **Clinton Enwerem**, John S. Baras, and Calin Belta. EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows, June 2026. arXiV Link. *2026*

Refereed Conference Publications

[C6] **Clinton Enwerem**, John S. Baras, and Calin Belta. Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty, December 2026. *To appear in the Proceedings of the 65th IEEE Conference on Decision and Control (CDC).* arXiV Link. *2026*

[C5] **Clinton Enwerem**, Shreya Kalyanaraman, John S. Baras, and Calin Belta. Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty. *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3117–3124, Pittsburgh, PA, USA, September 2026. arXiV Link. *2026*

[C4] **Clinton Enwerem**, Aniruddh G. Puranic, John S. Baras, and Calin Belta. Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression. *2025 IEEE 64th Conference on Decision and Control (CDC)*, pages 4890–4895, December 2025. arXiV Link. *2025*

[C3] **Clinton Enwerem**, Erfaun Noorani, John S. Baras, and Brian M. Sadler. Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling. *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 1087–1094, December 2024. arXiV Link. *2024*

[C2] **Clinton Enwerem** and John S. Baras. Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions. *2024 European Control Conference (ECC)*, pages 3762–3768, June 2024. *2024*

[C1] **Clinton Enwerem** and John S. Baras. Consensus-Based Leader-Follower Formation Tracking for Control-Affine Nonlinear Multiagent Systems. *9th International Conference on Control, Decision and Information Technologies (CoDIT)*, July 2023. *2023*

Refereed Journal Articles

[J1] **Clinton Enwerem** and John S. Baras. Formation Tracking for a Class of Uncertain Multiagent Systems: A Distributed Kalman Filtering Approach. *IEEE Control Systems Letters*, volume 8, 2024. *2024*

Research Experience

Institute for Systems Research, University of Maryland 8/2021–Present
Graduate Research Assistant *College Park, MD*

Research Topics: *Dexterous Manipulation, Robot Learning, Motion Planning, State Estimation, and Feedback Control*

- Trained PPO with grasp-distance-field rewards in **1,024 parallel Isaac Lab environments**; a Unitree G1 with a dexterous hand reached the pregrasp pose in **64/64** evaluation episodes versus **0/64** with palm-distance rewards.
- Fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing; each of three training seeds achieved 136/200 task completions versus 129/200 without supervision, recovering seven failures without disrupting nominal successes.
- Developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. Evaluated task execution in MuJoCo.
- Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under friction, pose, and contact uncertainty.
- Derived FIRMGrasp, a CVaR wrench-space margin for grasp selection under friction uncertainty, and implemented Grasp Distance Field control for feedback execution in simulated arm–hand systems.
- Built ParcelStow, an Isaac Lab benchmark for assessing imitation-policy robustness to execution-timing variation; generated **970K** expert states (**297** demonstrations) and trained ACT, Diffusion Policy, and DAgger. Controlled timing stress tests exposed precision-insertion failures: at 2× execution speed, ACT achieved **53%** success versus **84%** for the expert.
- Developed dexreach, extending cuRobo with GPU-accelerated dexterous grasp synthesis, friction and actuator feasibility checks, whole-body motion planning, and Isaac Lab sequence validation across arm–hand and humanoid models.
- Released ROS 2 manipulation software, including realhand_l6_ros2 for tactile dexterous-hand control over SocketCAN and kinematic_planner_ros2 for sampling-based manipulator planning with FCL collision checking.
- Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with Cartesian velocity control, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
- Published first-author work at CoDIT, ECC, CDC, and IEEE L-CSS, with additional papers to appear at IROS 2026 and CDC 2026 and preprints on manipulation and robot learning.

Electrical Engineering Department, University of Nigeria 9/2018–3/2021
Research Assistant *Enugu, Nigeria*

Research Topics: *Robust Control, Observer-Based Compensator Design, Feedback Linearization*

- Designed and implemented robust nonlinear control, active disturbance rejection, feedback linearization, and observer-based compensation for hardware motor control, released as open-source software, ADRC-Motor-Control.
- Co-authored the resulting peer-reviewed journal paper.

EE Department, University of Nigeria 8/2017–10/2017
Undergraduate Research Assistant *Enugu, Nigeria*

Research Topics: *Feedback Control, Time-Delayed Systems, System Identification*

- Designed feedback controllers for first-order-plus-dead-time processes with dead-time compensation and closed-loop system identification, with the implementation and write-up released as foptd-pade.

Professional Experience

Summer Research Assistant Jun.–Aug. 2023
Institute for Systems Research, University of Maryland | College Park, MD

Supervisor: *Professor John S. Baras*

- Formulated chance-constrained control for multi-agent systems with competing safety constraints.
- Derived Boolean compositions of control barrier functions and evaluated the controller in simulation; the resulting work was published at the 2024 European Control Conference.

MATRIX Lab, USM at Southern Maryland Jun.–Aug. 2022
Research Intern *California, MD*

Supervisor: *Dr. Danilo Romero*

- Conducted system-identification experiments for a Crazyflie quadrotor model.
- Implemented a Lagrangian optimal-control method in ROS-compatible Python for formation tracking with 10 Crazyflie quadrotors under localization uncertainty.

Kognitive Robotics 3/2021–8/2021
Robotics Engineer *Lagos, Nigeria*

Kognitive Robotics is a robotics startup developing turnkey robotics education platforms.

- Designed power and sensor–actuator interfaces in Altium; assembled the mobile robot and integrated its sensors and actuators.
- Implemented the ROS software stack and tested hardware–software integration on the physical robot.

Robotics & Artificial Intelligence Nigeria (RAIN) 1/2020–2/2021
Robotics Trainee *Ibadan, Nigeria*
- Built and tested robotics and IoT prototypes spanning mechanical design, sensor fusion, control firmware, and embedded software.
- Developed a ROS-based quadrotor autonomy stack on a Raspberry Pi 3, integrating ORB-SLAM3 visual SLAM, an EKF fusing SLAM poses with IMU data, and a PID flight controller. The onboard controller maintained stable hover during ground-station packet loss.

Teaching Experience

ECE Department, University of Maryland 9/2025–12/2025
Graduate Teaching Assistant *College Park, MD*
Course: *ENEE467: Robotics Project Laboratory* (with Prof. Calin Belta)
- Mentored 50+ undergraduate students in kinematics, 3D perception, motion planning, and ROS 2 control.
- Built reproducible Docker-based labs and testing pipelines.
- Led weekly lab sessions and provided technical feedback on documentation and debugging.

ECE Department, University of Maryland 2/2025–5/2025
Graduate Teaching Fellow *College Park, MD*
Course: *ENEE661: Nonlinear Control Systems* (with Prof. John S. Baras)
- Led problem-solving sessions on nonlinear analysis and control, covering phase-plane and bifurcation analysis, and stability concepts including Lyapunov, input-to-state, L_2, and L_∞ stability.

Technical Skills

Robotics: cuRobo, ROS/ROS2, ros2_control, MoveIt 2, MoveIt Servo, VR teleoperation, DimOS, MuJoCo, Drake, Gazebo, Isaac Sim, Isaac Lab, Pinocchio, RViz, URDF/SRDF/MJCF, SocketCAN; **Motion Planning:** RRT*, Informed RRT*, Sampling-Based Planning, FCL Collision Checking, OMPL; **Control & Safe Learning:** Safe & Constrained Reinforcement Learning, Control Barrier Functions, Safety Filters, Constrained MDPs, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Lyapunov Stability; **Robot Learning & Generative Models:** Vision-Language-Action (VLA) Models, OpenVLA-7B, QLoRA, Imitation Learning & Behavior Cloning, ACT (Action Chunking with Transformers), DAgger, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML:** PyTorch, TensorFlow; **Perception & State Estimation:** FoundationPose, BundleSDF, SAM 2, SAM 3D, PCL, OpenCV; **Testing & Validation:** mock hardware, virtual CAN, launch_testing, gtest/pytest, CI; **GPU Computing:** CUDA, Warp, MuJoCo Warp, JAX; **Programming:** Python, C++, MATLAB, Bash, LaTeX; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi, Mosek, Pyomo; **RL Sandboxes:** OpenAI Gym, Safety Gymnasium, PyBullet; **Dev Tools:** git, GitHub, GitLab, Docker; **Robots & EoATs:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, Unitree G1 (simulation), LEAP Hand v1, RealHand L6, RealHand A7 (Bimanual Arm-Hand System).

Honors & Awards

2024: IEEE CSS Student Travel and Workshop Support: Conference travel award to attend CDC'24.
2022: '22–'23 Microsoft Diversity in Robotics & Autonomy PhD Fellowship, MRC & Microsoft.
2022: ROSCon Diversity Scholarship: Travel grant to attend ROSCon 2022 in Kyoto, Japan.
2021: Finalist, Engineers' League, Pan-African Robotics Competition, Rwanda.
2021: CIT Dean's Fellowship, Carnegie Mellon University, Africa Campus, Kigali, Rwanda.
2021: Dean's Fellowship, University of Maryland, College Park, MD, United States.
2020: Scholar, Stanford Exposure to Research & Graduate Education, Stanford University, CA, USA.
2020: EducationUSA Opportunity Funds Program Scholarship, U.S. Consulate General, Lagos, Nigeria.
2020: Sole Recipient (Nationwide), Door Foundation Leadlight Scholarship, RAIN.
2016–2018: Agbami Science & Technology Scholarship, Chevron: Merit-based undergraduate scholarship.
2015–2018: MTN Foundation Scholarship: Nationwide merit-based undergraduate scholarship.

Professional Training & Development

Certificate in Robot Development & Automation 1/2020–2/2021
Robotics & Artificial Intelligence Nigeria (RAIN)
Completed graduate-level coursework and projects in robotics, control theory, ML, and IoT.

Open Courseware Summer 2022, 2023, 2025; Fall 2025
Bayesian Statistics (*UCSD, Coursera*), Autonomous Navigation for Flying Robots (*TUM, edX*), Principles of Robot Autonomy I & II (*Stanford*), Planning & Learning in Robotics (*UCSD*), Robot Learning (*USC*).

Professional Service, Outreach, & Mentoring

2023–2026: *Peer Reviewer*: Heliyon, MED'23, ECC'24, ACC'25, L4DC'25, CDC'25, CDC'26, ICRA'27, IEEE Open Journal of Control Systems, Elsevier (Robotics & Autonomous Systems).
2021–Present: *Member*, Black in Robotics (BiR): advancing Black representation in academic and industry robotics.
2021–2022: *Mentor*, EducationUSA & iScholar Initiative: Helped underrepresented STEM graduates secure fully funded admission to Stanford, Purdue, and Penn State.